Class Activity CO3

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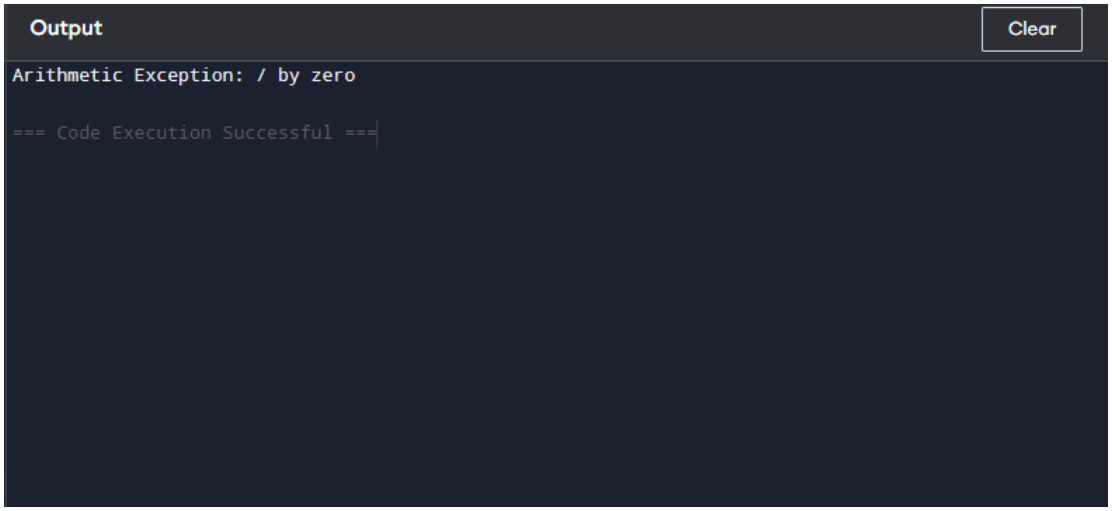
**1. Multiple Catch Statements**

```
public class Main {
   public static void main(String[] args) {
      try {
         int a = 10 / 0;
         int[] x = {1, 2};
         System.out.println(x[5]);
      }
      catch (ArithmeticException e) {
         System.out.println("Arithmetic Exception: " + e.getMessage());
      }
      catch (ArrayIndexOutOfBoundsException e) {
         System.out.println("Array Exception: " + e.getMessage());
      }
   }
}
```

**Output:**

Arithmetic Exception: / by zero

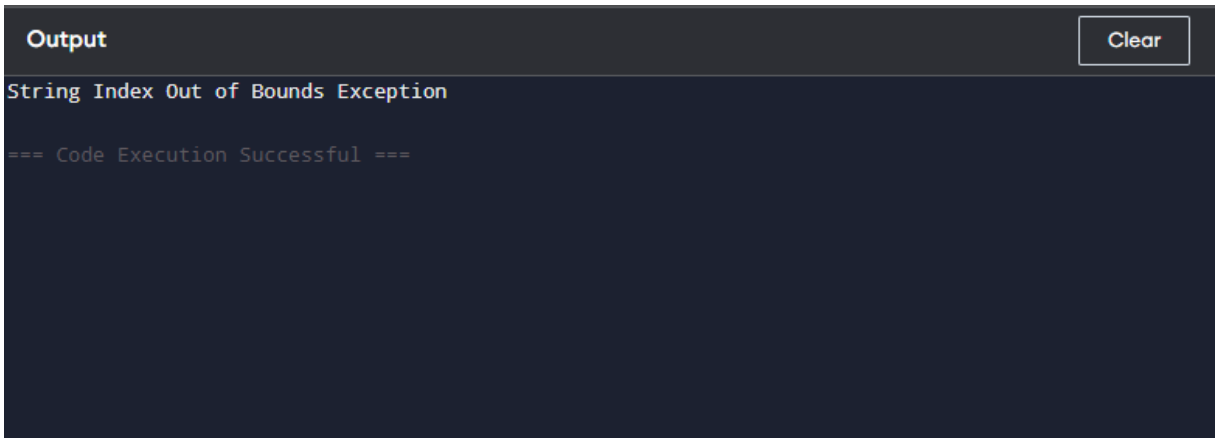


**2. StringIndexOutOfBoundsException**

```
public class Main {
    public static void main(String[] args) {
        String s = "Java";

        try {
            System.out.println(s.charAt(10));
        }
        catch (StringIndexOutOfBoundsException e) {
            System.out.println("String Index Out of Bounds Exception");
        }
    }
}
```

**Output:**

String Index Out of Bounds Exception



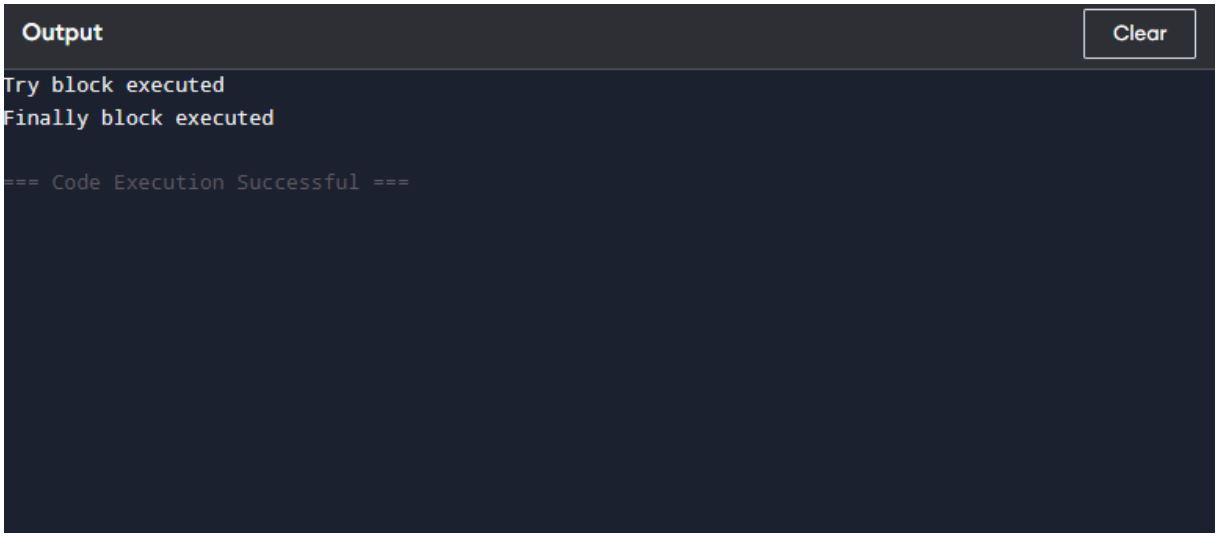
**3. finally**

```
public class Main {
    public static void main(String[] args) {
        try {
            System.out.println("Try block executed");
        }
        finally {
            System.out.println("Finally block executed");
        }
    }
}
```

**Output:**

Try block executed

Finally block executed

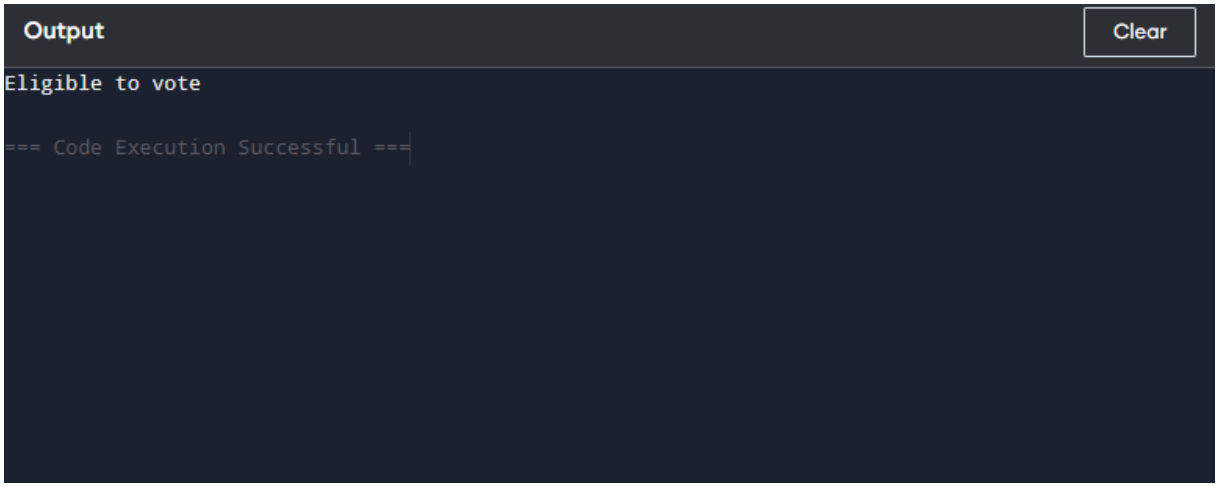


**4. throw – Voting Eligibility**

```
public class Main {

    static void checkAge(int age) {
        if (age < 18)
            throw new ArithmeticException("Not eligible to vote");

        System.out.println("Eligible to vote");
    }

    public static void main(String[] args) {
        try {
            checkAge(20);
        }
        catch (ArithmeticException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

**Output:**

Eligible to vote



**5. throws**

```
import java.io.*;

public class Main {

    static void readFile() throws IOException {
        FileReader f = new FileReader("sample.txt");
        f.close();
    }

    public static void main(String[] args) {
        try {
            readFile();
            System.out.println("File opened successfully");
        }
        catch (IOException e) {
            System.out.println("IOException: " + e.getMessage());
        }
    }
}
```

**Output:**

IOException: sample.txt (No such file or directory)